

**A SEMINAR REPORT ON  
RASPBERRY PI BASED WEATHER  
REPORTING OVER IOT**

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# CERTIFICATE

This is to certify that the Seminar report entitled  
**RASPBERRY PI BASED WEATHER REPORTING  
OVER IOT**

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is a bonafide work carried out by them under the supervision of Dr.Dhaigude N.B.  
and it is approved for the partial fulfillment of the requirement of Savitribai Phule  
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This Seminar report has not been earlier submitted to any other Institute or Uni-  
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# Abstract

In this proposed is an advanced solution for monitoring the weather conditions at a particular place and make the information visible anywhere in the world. The technology behind this is Internet of Things (IoT), which is an advanced and efficient solution for connecting the things to the internet and to connect the entire world of things in a network Here things might be whatever like electronic gadgets, sensors and automotive electronic equipment. The system deals with monitoring and controlling the environmental conditions like temperature, relative humidity, light intensity and CO level with sensors and sends the information to the web page and then plot the sensor data as graphical statistics. The data updated from the implemented system can be accessible in the internet from anywhere in the world.

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# List of Symbols, Abbreviations and Nomenclature

| Symbol | Details                              |
|--------|--------------------------------------|
| IoT    | Internet of Things                   |
| IDE    | Integrated Development Environment   |
| GPIO   | General Purpose Input Output         |
| ADC    | Analog to Digital Converter          |
| HDMI   | High Definition Multimedia Interface |
| USB    | Universal Serial Bus                 |
| LCD    | Liquid Crystal Display               |

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# Chapter 1

## Preamble

### 1.1 Introduction

Weather or Climate is important part of human life. Sensors are essential components not only applicable to the industries for process control but also in daily life for safety of building's and traffic flow measuring, environmental parameters measurement. IoT means Internet of Things. It provides internetworking of physical devices, buildings, vehicles and other components like sensors and actuators. By giving network connectivity to systems embedded with electronics, software, sensors and actuators; these objects are able to collect and exchange data. By using IoT objects to be sensed or controlled remotely through existing network. It gives opportunity to connect physical world with computer-based systems. IoT improves efficiency, accuracy, economic benefits along with reduced manpower. IoT frameworks help for the interaction between "things". Also supports for more complex structures like distributed computing and development of distributed applications. Now a days most of IoT frameworks seem to focus on real-time data logging solutions. The data of the measured parameters are not useful if they are not transmitted fast and accurate manner to the users. Therefore, transmitted and processing the measured data is a very important aspect of the modern weather forecast. Transmission of the measured data could be done by anumber of means: WI-FI link, GSM/GPRS link, satellite link direct, wired link, etc. Weather forecasting has to be reliable and accurate, regardless of its application. Also, it has to provide simple access to all the measured parameters. The quality of sensors



and precision of measurements may vary, and the location of weather forecasting station can determine the accuracy and reliability of the weather data collection. Raspberry Pi, acting as data logger process the converted output of sensors from analog to digital. The logged data can then be transferred to a desktop or any other monitor has GUI for further analysis. So by using easily obtained components and less complicated circuitry powerful weather station can be build Now a day's various weather factors like wind and many other cause great impact on humans day to day life.

## 1.2 Problem Statement

Existing weather reporting systems primarily rely on centralized meteorological stations that may not provide localized data. This can result in a lack of awareness about immediate weather changes, which can adversely affect farmers, outdoor workers, and communities vulnerable to severe weather events.

## 1.3 Objectives

1. Collects real-time environmental data (temperature, humidity, atmospheric pressure, and rainfall).
2. Transmits data to a cloud server for storage and analysis.
3. Provides users with accessible, real-time weather updates through a web interface or mobile application.
4. Allows users to receive alerts for severe weather conditions.

## 1.4 Literature Survey

The literature survey by Bhagat AM, Thakare AG, KAM, Muneshwar NS, and Choudhary PV examines recent advancements. It synthesizes key findings from various studies, highlighting methodologies, results, and emerging trends. The authors identify critical gaps in the existing research, emphasizing the need for further

exploration in areas. This survey serves as a valuable resource for researchers, guiding future investigations and fostering a deeper understanding of the subject. Describes how the literature was collected and analyzed, including databases searched and inclusion/exclusion criteria. By [1]

The literature survey by U. Hariharan and K. Rajkumar reviews recent developments in. It summarizes key research findings, methodologies, and trends while highlighting significant contributions from various studies. The authors identify gaps in the current literature, underscoring areas that require further investigation. By providing a comprehensive overview, this survey serves as a foundational resource for researchers and practitioners, facilitating informed discussions and guiding future research efforts. By [2]

Joseph FJJ's literature survey explores recent advancements. It provides a critical analysis of key studies, highlighting methodologies, findings, and emerging trends. The survey identifies significant gaps in the existing research, suggesting areas that warrant further exploration. By synthesizing a diverse range of sources, this work serves as an essential reference for researchers, offering insights that can guide future studies and enhance understanding. By [3]

Muck PY and Homam MJ's 2018 literature survey provides a comprehensive analysis of advancements. It synthesizes key findings from various studies, focusing on methodologies and emerging trends. The authors highlight critical gaps in the literature, emphasizing areas for further research and exploration. This survey serves as an important resource for researchers and practitioners, facilitating a deeper understanding and guiding future investigations in the field. By [4]

The literature survey by S. S. Sengar, U. Hariharan, and K. Rajkumar examines significant developments. It synthesizes key research findings, highlighting diverse methodologies and critical trends. The authors identify notable gaps in the current literature, suggesting avenues for future investigation. By providing a thorough overview of existing studies, this survey serves as a valuable reference for researchers and practitioners, fostering a deeper understanding and guiding subse-

quent research efforts in the area. By [5]

The literature survey by Adhav Gaurav, Jadhav Kunal, and Nadlamani Kavita (2016) reviews key advancements. It summarizes significant research findings, methodologies, and trends, offering insights into the evolution of the topic. The authors highlight critical gaps in existing literature, advocating for further exploration in under-researched areas. This comprehensive overview serves as an essential resource for researchers and practitioners, enhancing understanding and informing future research directions in the field. By [6]

## Chapter 2

# Block Diagram and Its Explanation

### 2.1 Block Diagram

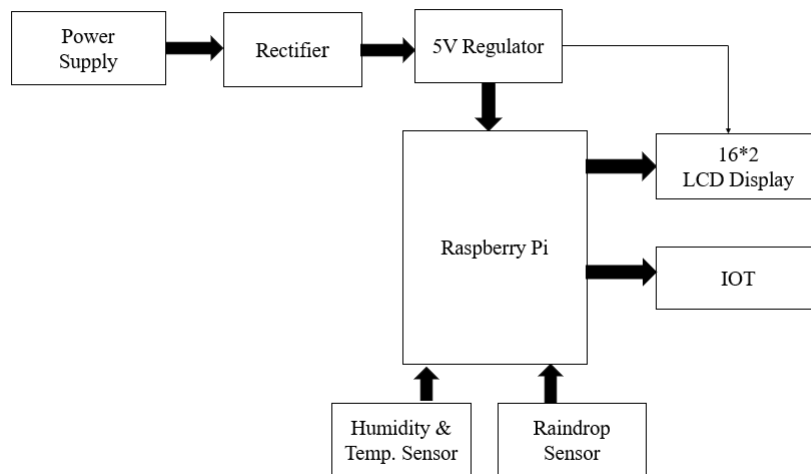


Figure 2.1: Block Diagram

Raspberry pi is the latest wireless technology .Proposed System will visualize and store various weather parameters as given above with the help of sensors interfaced to Raspberry will get all data, SD card on Pi stores the collected data as like memory card. Then at the output side LCD is to be connected for showing

the result and on off relays for server access. Also 5V, 1A power supply is given to the Raspberry Pi board through micro USB slot. An SD card of 8GB is used to store the operating system as well as all programs and files needed for this project. Keyboard and mouse is connected to the Raspberry Pi board through USB ports. Monitor is connected to the Raspberry Pi board through HDMI port using HDMI to VGA cable. Ethernet port is used to give internet connection to the system via LAN. To know the current weather status at remote location, the user can to log in on web browser by entering username and password given for particular server by the user. Web application opens after entering password and with the output graphical representation also obtain. Raspberry pi processed data will updated continuously on cloud server user will get to know the stored data on hourly and daily basis. This item is the Raspberry Pi Model B, revision 2.0, which has 512 MB of RAM, an Ethernet port, HDMI output, RCA composite video output, audio output, two USB ports, and 0.1-spaced pins that provide access to general purpose inputs and outputs (GPIO). The Raspberry Pi requires an SD card with an operating system on it (not included). The Raspberry Pi is very popular, with lots of example projects and information available online.

# Chapter 3

## Component Used

### 1.Raspberry Pi

Raspberry PI is a card-sized ARM powered Linux computer development board.



Figure 3.1: Raspberry Pi

There are in total of 5 types of various board with different specification, for the proposed Weather forecasting system Raspberry PI 2 model B is used as the main development board which is shown in Figure-1. The raspberry pi consists of four USB Ports and one 10/100 Base T Ethernet Socket. Forty pins GPIO Header are

present in the raspberry pi board which is used for connecting to Analog to Digital converter chip (MCP3008) to which the sensors are connected. A 5V Micro USB power port is present to which the power supply is given for the device. A HDMI port is present through which interfacing of the monitor and the Raspberry pi can be done and the USB ports for the keyboard and mouse interfacing. At the bottom a Micro SD Card Slot is provided where the Micro SD card is to be inserted with the Raspbian Jessie botting software which based on the Linux platform. The GPIO pins have different uses individually such as power supply, ground, clock, UAR

2. **DHT11 Sensor:**

Measurement and control of temperature and relative humidity finds applications

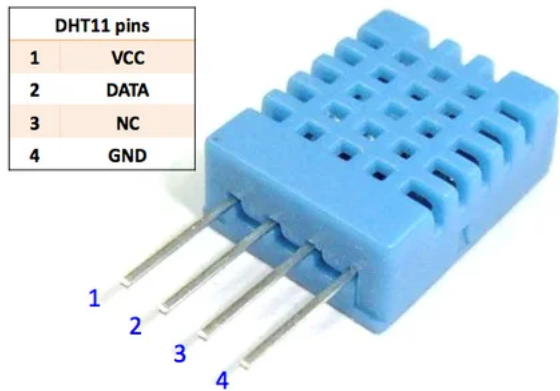


Figure 3.2: DHT 11 Sensor

in numerous areas. These days devices are available which have both temperature and humidity sensors with signal conditioning, ADC, calibration and communication interface all built inside them. The use of such smart sensors greatly simplify the design and reduces the overall cost. Humidity and temperature measurements with Sensirion’s SHT1x/SHT7x sensors. These sensors are capable of measuring both temperature and relative humidity and provide fully calibrated digital out-

puts. While SHT1x/SHT7x are very accurate sensors, they are still expensive for hobbyists use. This articles discusses the DHT11 sensor which also provides calibrated digital outputs for temperature and humidity but is relatively lot cheaper than the Sensirion sensors. The 19 DHT11 sensor uses a proprietary 1-wire protocol which we will be exploring here and implementing with the Raspberry Pi that will receive the temperature and humidity values from the sensor and display them on cloud.

### 3. Raindrop Sensor

A raindrop sensor is an electronic device used to detect the presence of rain or

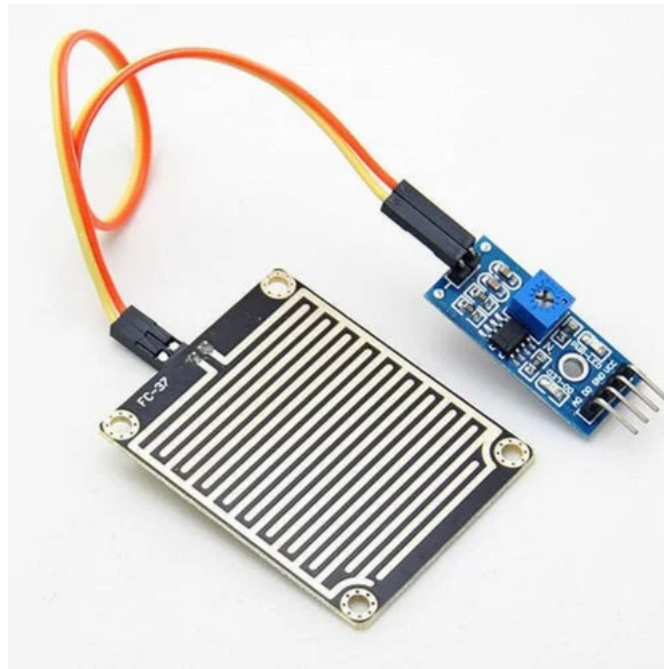


Figure 3.3: Raindrop Sensor

water droplets. It typically consists of a rain detection module with a conductive plate or a grid that becomes conductive when wet, allowing it to sense moisture. The sensor works by detecting changes in conductivity, capacitance, or resistance when water droplets land on the plate. When rain is detected, the sensor can trigger various actions, such as closing windows, turning off sprinklers, or providing



input for weather monitoring systems. Raindrop sensors are commonly used in automated irrigation, smart homes, and weather stations.

4. **LCD Display** A 16x2 LCD display is a widely used alphanumeric display



Figure 3.4: LCD Display

module that can show up to 16 characters on two rows, making a total of 32 characters. Each character is displayed in a 5x7 or 5x10 pixel matrix, providing clear visibility of letters, numbers, and symbols. The display operates on a parallel interface, often using the HD44780 driver, which allows simple integration with microcontrollers like Arduino, Raspberry Pi, or PIC. It supports ASCII characters and some custom symbols, which can be programmed as needed.

This type of LCD is commonly used in electronic projects due to its affordability, ease of use, and low power consumption. It also features an optional backlight, enhancing visibility in low-light environments. Applications range from displaying sensor data, system statuses, or user inputs in home automation, robotics.

# Chapter 4

## Circuit Diagram

### 4.1 Circuit Diagram

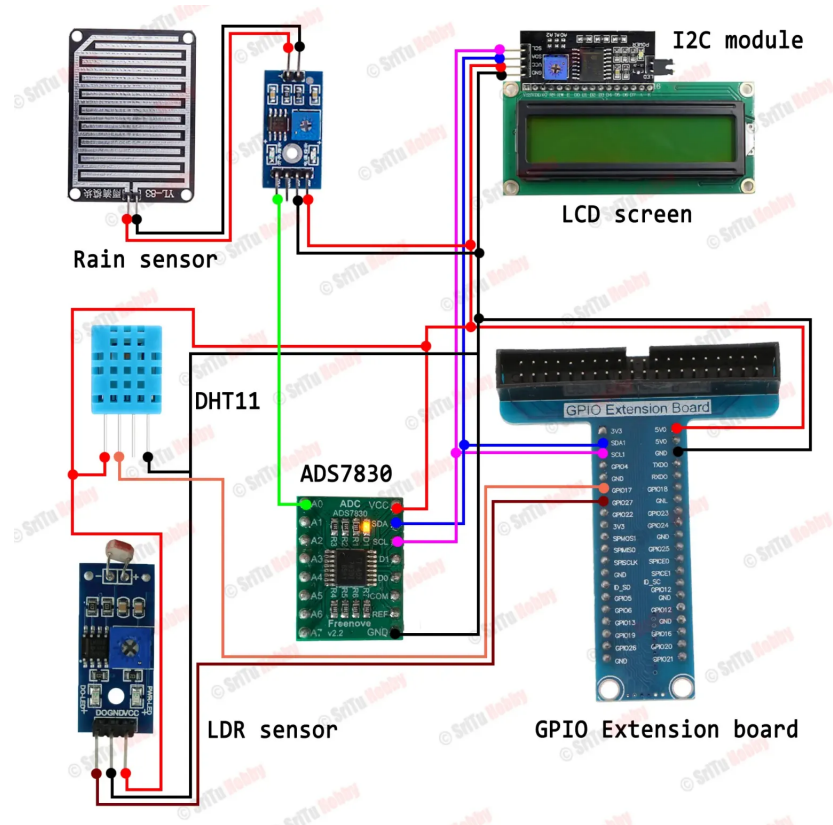


Figure 4.1: Circuit Diagram

This Raspberry Pi-based weather monitoring system uses IoT technology to

track environmental parameters like temperature, humidity, and rainfall. The system integrates sensors such as the DHT11 and a raindrop sensor. The DHT11 is responsible for measuring temperature and humidity, while the raindrop sensor detects rainfall intensity. These sensors are connected to the Raspberry Pi via its GPIO (General Purpose Input/Output) pins.

**Circuit Overview:**

1. **DHT11 Sensor:** The DHT11 has three pins: VCC, Ground, and Data. The VCC pin connects to the 5V pin on the Raspberry Pi, Ground to a common ground, and Data to one of the GPIO pins to read temperature and humidity data.
2. **Raindrop Sensor:** The raindrop sensor provides an analog output, which must be converted to digital using an ADC (such as MCP3008). This allows the Raspberry Pi to process the rainfall data. The sensor's VCC and Ground connect to 5V and Ground on the Raspberry Pi, while the output pin connects to the ADC, and then to the GPIO pins for reading data.
3. **LCD Display:** The LCD is connected to the Raspberry Pi using the I2C interface, which simplifies wiring and uses fewer pins compared to direct connections. It displays real-time weather data from the sensors.

The Raspberry Pi processes the sensor data and displays it on the LCD. The IoT functionality allows the data to be transmitted wirelessly using protocols such as MQTT or HTTP, enabling remote monitoring through a web-based interface or mobile app, providing an efficient and scalable weather monitoring solution.

# Chapter 5

## Algorithm

Here's the algorithm for the Raspberry Pi-based weather monitoring system using DHT11, raindrop sensor, and an LCD display:

**1.Start:** Initialize the Raspberry Pi and set up the required libraries.

**2.Initialize Sensors:** - Set up the GPIO pins for the DHT11 temperature and humidity sensor. - Initialize the ADC (like MCP3008) for the raindrop sensor to convert the analog signal to digital. - Initialize the I2C communication for the LCD display.

**3.Read Data from Sensors:** - Read temperature and humidity values from the DHT11 sensor. - Read the analog signal from the raindrop sensor, convert it to digital using the ADC, and determine rainfall status.

**4.Process Data:** - Store the temperature, humidity, and rainfall data. - Format the sensor data for display (e.g., convert raw values into understandable units like °C for temperature,

**5.Display Data on LCD:** - Send the formatted temperature, humidity, and rainfall data to the LCD display using I2C communication.

**6.Send Data to Cloud (Optional):** - If IoT integration is required, send the sensor data to a remote server or dashboard using MQTT/HTTP protocols for remote monitoring.

**7.Repeat:** - Continuously loop the data reading, processing, and displaying steps at regular intervals

**8.End:** The system runs indefinitely unless manually stopped or a failure.

# Chapter 6

## Flowchart

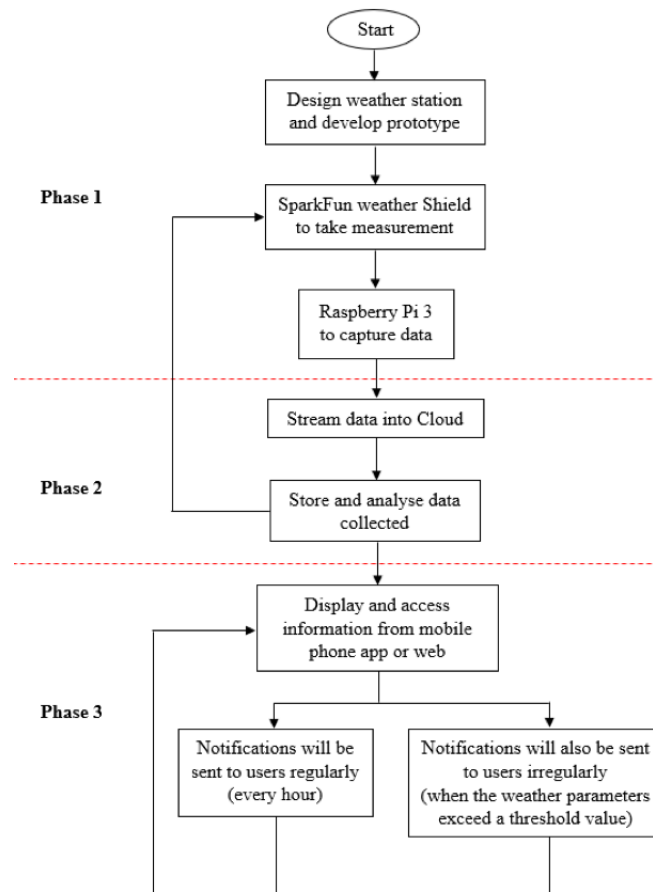


Figure 6.1: Flowchart

# Chapter 7

## Result

Here is a detailed list of expected results for your Raspberry Pi-based weather reporting over IoT project:

1. **Real-time Data Acquisition:** The sensors connected to the Raspberry Pi (such as temperature, humidity, and pressure sensors) will continuously collect weather-related data in real time.
2. **Data Transmission:** The Raspberry Pi will transmit the acquired weather data to a cloud server or IoT platform over a network (Wi-Fi, Ethernet, or cellular).
3. **Cloud Storage:** Weather data will be stored securely on a cloud-based platform or database (such as Firebase, AWS IoT, or ThingSpeak) for remote access and long-term analysis.
4. **IoT Dashboard:** A user-friendly IoT dashboard or web application will display the weather data in graphical and tabular formats, making it easy to visualize trends and monitor conditions remotely.

# Chapter 8

## Conclusion

The design and development weather monitoring system is very challenging this system has been tested successfully in different areas with various terrain, in various conditions. The performance and efficiency of weather monitoring is increased by the help of this Raspberry Pi. This Raspberry Pi has a good capturing capability, so that the Big number values are easily manipulated, because of this the quality of values increased. This can be used throughout day and night which reduces the labor cost. Since it was designed to be user friendly, the operator can control the Weather monitoring system easily .In this paper, a comparative analysis on various systems developed for the controlling and monitoring system for the industry is proposed. Design and implementation of the existing system is also discussed. After reviewing the existing studies, a cost efficient system has been proposed for industrial monitoring and controlling based on IoT framework. In our system various parameters are continuously monitored remotely that to without any human interference. The system ensures a safety work environment for the worker. Facial recognition based logging system is designed to prevent the network intrusion.

# Chapter 9

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